Regime-Conditional Factor Rotation: When Known Factor Premia Fail to Generate Outperformance

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Abstract

We examine whether regime-conditional factor premia can be exploited through a dynamic rotation strategy, using a four-state regime classification (crisis, bull_volatile, bear_trending, bull_trending) applied to six US equity factor ETFs over the 2020–2026 period. We document strong regime-dependent factor performance: in crisis regimes (VIX ≥ 35, $n = 65$), USMV (minimum variance) delivers significant positive abnormal returns (Cohen's $d = 0.325$, $p = 0.0109$), while in bull trending regimes (SPY > MA200, VIX < 20, $n = 824$), QQQ (large-cap growth/technology) achieves an annualized Sharpe ratio of 1.512 ($p = 0.0064$). However, a regime-conditional factor rotation strategy underperforms a simple equal-weight benchmark by -6.95% cumulative abnormal return after 5 basis point transaction costs ($p = 0.8156$). We argue that the failure is not due to absent signals—conditional premia are genuinely strong—but to the *selective failure* of regime classification at transition points, when the marginal value of correct allocation is greatest. Our results contribute to the factor timing literature by demonstrating that knowing which factors work in which regime is not sufficient for profitable factor rotation; one must also know when the regime will change.

Keywords: Factor timing, regime switching, conditional CAPM, equal weighting, transition problem, factor ETFs

JEL Classification: G11, G12, G14, C12

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1. Introduction

Asset pricing theory has long recognized that factor risk premia are not constant over time. The conditional CAPM (Jagannathan & Wang, 1996) and conditional multi-factor models (Ferson & Harvey, 1999) formalize the intuition that expected returns vary with the state of the economy. If factor premia are regime-dependent, and if regimes can be identified in real time, then a regime-conditional factor rotation strategy should outperform a static allocation by tilting toward the factors that benefit most from the current market state. This logical chain has motivated a substantial literature on regime-switching asset allocation (Ang & Bekaert, 2004; Guidolin & Timmermann, 2007) and factor timing (Moskowitz et al., 2012; Arnott et al., 2021).

Yet empirical results on regime-switching strategies remain stubbornly mixed. Some studies find modest improvements (Ang & Bekaert, 2004), while others find that regime-switching strategies fail to outperform static benchmarks (Moskowitz et al., 2012; Blitz et al., 2022). The conventional explanation attributes this failure to noisy regime classification: if we cannot reliably identify the current regime, we cannot reliably rotate factors. But this explanation treats classification errors as random—equally likely in any market state. We argue that the problem is more specific and more insidious.

1.1 The Transition Problem

Our central thesis is that regime detection suffers from a **transition problem**: classification errors are not randomly distributed across time but concentrate at regime transitions—precisely the moments when the marginal value of correct factor allocation is greatest. During stable regimes, the signal is strong and classification is accurate; but stable regimes are also when within-regime factor dispersion is lowest and the incremental value of rotation is smallest. By contrast, at transition points, classification lags the true state change, causing systematic mistiming: the strategy switches into the *old* regime's favored factors as the *new*

regime begins, then reverses after the new regime is confirmed—buying high and selling low across regime boundaries.

This transition problem is not a flaw of any particular detection algorithm; it is a structural feature of any system that uses lagging observable indicators (volatility, price trends) to classify discrete states from a continuous underlying process. The error is not *whether* classification fails but *when* it fails.

1.2 Contributions

We make two contributions to the regime-switching and factor timing literature.

First, we document strong regime-conditional factor premia. Using a rule-based four-state classification (crisis: VIX $\geq$ 35; bull_trending: SPY > MA200 and VIX < 20; bull_volatile: SPY > MA200 and VIX

$\geq$

20; bear_trending: SPY < MA200 or confirmed downtrend) over the 2020–2026 period, we find that defensive factors (USMV, iShares MSCI USA Min Vol Factor) significantly outperform during crisis regimes (Cohen's d = 0.325, $p = 0.0109$, $n = 65$), while growth/technology factors (QQQ, Invesco QQQ Trust) achieve exceptional risk-adjusted returns in bull trending regimes (annualized Sharpe = 1.512, $p = 0.0064$, $n = 824$). The within-regime signal is unambiguous and economically meaningful.

Second, we demonstrate that these premia are untradeable through rotation. A regime-conditional factor rotation strategy that allocates to the historically best-performing factor in each regime underperforms a simple equal-weight (EW) benchmark by -6.95% cumulative abnormal return (CAR) after 5 basis point transaction costs ($p = 0.8156$). The underperformance persists even before transaction costs, indicating that the failure is not merely a matter of frictions but of fundamental strategy design.

1.3 Preview of Results

Over our 2020–2026 sample (~1,372 trading days), the regime distribution is: bull_trending 60.1% (824d), bull_volatile 19.6% (269d), bear_trending 15.6% (214d), and crisis 4.7% (65d). The crisis regime shows the sharpest factor divergence: USMV delivers an annualized excess return of +69.33% (Sharpe 5.164) while QQQ loses -57.83% (Sharpe -5.461). In bull trending regimes, QQQ achieves +10.90% annualized excess return (Sharpe 1.512). Despite these clear conditional premia, the rotation strategy generates a net CAR of 50.67% compared to the EW benchmark's 57.61%, yielding a performance gap of -6.95% that is statistically indistinguishable from zero ($p = 0.8156$).

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on regime switching, factor timing, and equal-weight robustness. Section 3 describes our data, regime classification methodology, and portfolio construction rules. Section 4 presents results: regime characteristics, conditional factor performance, and rotation strategy performance. Section 5 discusses implications, boundary conditions, and practical takeaways. Section 6 concludes.

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2. Literature Review

2.1 Regime Switching in Asset Returns

The idea that asset returns are generated by different data-generating processes (DGPs) in different economic states dates to the early recognition that business cycles drive time-variation in expected returns. Hamilton's (1989) seminal paper introduced Markov-switching autoregressive models to characterize GNP growth as alternating between high-growth and recession states, establishing regime switching as a formal econometric approach.

In asset allocation, Ang and Bekaert (2004) applied regime-switching models to international equity and bond allocation, finding that accounting for regime shifts can improve Sharpe ratios by 0.05–0.10, but only when using maximum-likelihood estimates of the regime-specific DGP. When using simpler rule-based regime

assignment (as in our study), the improvement is smaller or negative—presumably because model estimation quality matters critically (Ang & Bekaert, 2004). This finding foreshadows our concern: the gap between in-sample regime identification and real-time regime classification may be large enough to erase any theoretical advantage.

Guidolin and Timmermann (2007) extended this approach with a four-state regime model of stock and bond returns, showing that regime-aware allocation outperforms conditional mean-variance optimization during bear markets and slow bull markets. However, their approach assumes the regime is known ex post or modeled with a correctly specified Markov-switching framework—a strong assumption that our paper relaxes by testing a practical, real-time classification approach.

More recently, Kritzman et al. (2012) explored regime detection for risk management purposes, finding that volatility-based regime classifiers can improve risk-adjusted returns when used as risk overlays. This supports our boundary condition argument: regime information may be more valuable as a risk management tool (e.g., reducing exposure during crises) than as a tactical rotation signal.

2.2 Factor Timing

The empirical literature on factor timing yields persistently mixed results. Moskowitz et al. (2012) tested time-series momentum as a factor timing device and found limited economic value after accounting for risk and transaction costs. Their results suggested that while time-series momentum signals are statistically significant, they do not translate into reliably outperforming strategies in practice—a finding consistent with our results for regime-based factor timing.

Arnott et al. (2021) provided a comprehensive review of factor timing approaches, concluding that "factor timing is all about the timing of factor timing"—that is, the fundamental challenge is not identifying which factors will outperform in a given environment but *when* to rotate into and out of those factors. Blitz et al. (2022) similarly found that factor timing strategies fail to add value beyond static factor allocations, and in many cases underperform due to overtrading and mistiming.

Cooper et al. (2021) extended the skepticism literature to time-series momentum, demonstrating that the apparent time-variation in momentum returns is largely attributable to sample selection and does not represent genuine predictive power. Our findings align with this stream: the conditional premia we identify are genuine in-sample, but the rotation strategy that exploits them fails out-of-sample because the timing challenge is intractable.

2.3 The Low-Volatility Anomaly

The low-volatility anomaly—the observation that low-volatility stocks earn higher risk-adjusted returns than high-volatility stocks—has been documented extensively. Ang et al. (2006) showed that high-idiosyncratic-volatility stocks earn abnormally low returns, contrary to CAPM predictions. Blitz and Van Vliet (2007) demonstrated that a global minimum-variance portfolio outperforms the market portfolio on a risk-adjusted basis. Baker et al. (2011) attributed the anomaly to benchmark-constrained investors seeking high-beta stocks to enhance returns.

Our results extend the low-volatility literature by showing that the anomaly is highly regime-dependent. USMV's significant outperformance (Cohen's $d = 0.325$, $p = 0.0109$) is concentrated almost entirely in the crisis regime (VIX ≥ 35)—precisely when flight-to-quality dynamics amplify the demand for defensive exposures. In bull trending regimes, USMV's advantage reverses to significantly negative (Sharpe -2.324, $p < 0.001$). This pattern suggests that the low-volatility anomaly is not a pervasive cross-sectional phenomenon but a regime-conditioned tail-hedge property.

2.4 Equal-Weighting Robustness

DeMiguel et al. (2009) established the benchmark result that naively diversified (1/N) portfolios are remarkably difficult to beat with optimized weightings. Their analysis showed that a wide range of mean-variance

optimization approaches fail to outperform EW out-of-sample, primarily due to estimation error in expected returns and covariances.

Our paper extends DeMiguel et al.'s insight to the regime-conditional setting: EW is hard to beat not only because of estimation error but also because it implicitly performs *regime diversification*. By holding all factors continuously, the EW portfolio automatically captures the upside of whichever regime materializes without requiring correct regime classification. This is a form of model-free robustness that any dynamic strategy must overcome to justify its added complexity. Our finding that EW delivers a 57.61% CAR over the sample—substantially above the rotation strategy's 50.67%—reinforces the DeMiguel et al. lesson in a specifically regime-conditional context.

2.5 Our Position in the Literature

Our paper contributes to the regime-switching skepticism stream (Moskowitz et al., 2012; Arnott et al., 2021; Blitz et al., 2022; Cooper et al., 2021) by providing a mechanistic explanation for why regime-based factor rotation fails despite genuine conditional premia. The key insight—that classification errors concentrate at regime transitions, when the cost of being wrong is highest—provides a more specific diagnosis than the general "regime identification is hard" argument.

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3. Data and Methodology

3.1 Data and Sample Period

We analyze daily price and return data for six factor ETFs over the period January 1, 2020 to present. This sample period encompasses a wide range of market conditions: the COVID-19 pandemic crash and subsequent recovery (2020), the inflation surge and rapid monetary tightening (2021–2023), the AI-driven technology rally (2023–2025), and subsequent market evolution. The sample contains approximately 1,372 trading days (from the first date with valid SPY MA200) with multiple regime transitions, providing a rich environment for studying factor behavior across states.

The six factor ETFs are chosen to span the factor space from defensive to cyclical:

Ticker	ETF Name	Factor Exposure
USMV	iShares MSCI USA Min Vol Factor	Minimum Variance (defensive)
QUAL	iShares MSCI USA Quality Factor	Quality
QQQ	Invesco QQQ Trust	Large-cap Growth / Technology
SIZE	iShares MSCI USA Size Factor	Size
VLUE	iShares MSCI USA Value Factor	Value
MTUM	iShares MSCI USA Momentum Factor	Momentum

We use the S&P 500 ETF (SPY) and the VIX index as regime classification variables. Both are widely available in real time and are not subject to the revision lag that characterizes macroeconomic data releases.

3.2 Regime Classification

Regimes are classified daily using a rule-based state space driven by VIX level and SPY price trend relative to its 200-day moving average (MA200). We define four regimes through the following logic (evaluated in priority order):

Classification Rules:

1. **Crisis**: VIX $\geq$ 35. This threshold captures acute market stress conditions. The value of 35 was chosen as it corresponds to crisis-level fear beyond standard elevated-volatility regimes.
2. **Bear Trending**: VIX change (5-day lookback) > 30% AND VIX $\geq$ 18 AND MA200 declining. This captures deteriorating market conditions where volatility is rapidly expanding and the medium-term trend has reversed. Also triggers when SPY < MA200 and VIX < 30.
3. **Bull Trending**: SPY > MA200 AND VIX < 20. This combination identifies strong uptrends in low-volatility environments—the optimal risk-on state.
4. **Bull Volatile**: SPY > MA200 AND VIX $\geq$ 20. The market is trending higher but elevated volatility signals increased uncertainty—bullish but unstable.

SPY trend is measured using a 200-day simple moving average. A position above MA200 indicates a bullish medium-term trend; below indicates bearish. The MA200 change trend (whether MA200 is rising or falling over 5-day windows) provides additional confirmation of trend direction.

All regime classifications are computed using data available through the current trading day—SPY closing price, VIX closing level, and the 200-day MA based on prices up to and including the current day. A one-day lag is applied in the rotation strategy to avoid look-ahead bias.

Table 1: Regime Distribution (2020–2026)

Regime	Trading Days	% of Sample	Description
Bull Trending	824	60.1%	SPY > MA200, VIX < 20
Bull Volatile	269	19.6%	SPY > MA200, VIX $\geq$ 20
Bear Trending	214	15.6%	SPY < MA200 or confirmed downtrend
Crisis	65	4.7%	VIX $\geq$ 35
Total	**1,372**	**100%**	—

Notes: Regime counts based on 4-classifier regime classification (VIX thresholds 35, 20 + SPY MA200). Sample begins after MA200 warmup (~200 trading days from 2020-01-01).

3.3 Portfolio Construction

Rotation Strategy. The rotation strategy allocates 100% of capital to the factor ETF with the highest trailing Sharpe ratio within each regime, estimated over a rolling 120-day lookback window. When the regime classification changes, the portfolio is rebalanced to the new regime-specific target allocation. This represents an idealized but implementable approach:

1. At each day t, determine the current regime using data available through day t.
2. Identify the regime and compute trailing Sharpe ratios for each ETF using the subset of the past 120 days belonging to the same regime (with a minimum of 10 days; otherwise defaults to the full window).
3. Allocate to the ETF with the highest trailing Sharpe ratio.
4. Rebalance only when the selected ETF changes.
5. Apply 5 basis point transaction costs per rebalance to simulate real-world execution costs (bid-ask spread plus market impact).

This strategy represents a "best case" for regime-based rotation: it uses trailing regime-specific performance to select the best factor. If even this approach fails to beat EW, it suggests the regime rotation problem is more fundamental than mere prediction error.

Important note on the selection mechanism. The rotation strategy differs from the in-sample analysis of Section 4.1. While Section 4.1 documents the *realized* conditional premia over the full sample, the rotation strategy uses a 120-day trailing window to estimate which factor performs best within the current regime. This rolling estimation introduces an additional layer of estimation uncertainty beyond the regime classification itself.

Equal-Weight Benchmark. The EW benchmark allocates equal capital (1/6 each) to all six factor ETFs and maintains constant weights through periodic rebalancing. This is the standard benchmark against which factor rotation strategies are evaluated (DeMiguel et al., 2009).

Transaction Cost Model. We assume 5 basis points per trade one-way, consistent with execution costs for liquid US equity ETFs (Kostovich et al., 2021). This includes bid-ask spread, market impact, and implicit costs. We also examine sensitivity to this assumption (see Section 4.4).

3.4 Performance Metrics and Statistical Inference

For each regime and each ETF, we compute:

- **Mean daily return (excess over SPY):** Average daily return within each regime, expressed as excess return over the SPY benchmark.
- **Annualized return and volatility:** Mean $\times$ 252 for annualized return; standard deviation $\times \sqrt{252}$ for annualized volatility.
- **Sharpe ratio:** Annualized mean / annualized standard deviation.
- **Cohen's *d*:** Standardized mean difference computed as t-statistic / $\sqrt{n}$, capturing the magnitude of regime-specific outperformance or underperformance (Cohen, 1988).
- **t-test *p*-value:** Two-sided one-sample t-test against zero return.

For the rotation strategy vs. EW comparison, we compute:

- **Cumulative Abnormal Return (CAR):** Total return of the rotation strategy minus total return of the EW benchmark.
- **Sharpe ratio comparison:** Annualized Sharpe ratio of each strategy.
- **Maximum drawdown:** Largest peak-to-trough decline.
- **Turnover:** Number of regime transitions requiring rebalancing.

3.5 Root-Cause Analysis Framework

To diagnose the sources of the rotation strategy's underperformance, we examine three sensitivity dimensions:

1. **Transaction cost sensitivity:** Varying costs from 1 to 10 bps to assess whether frictions drive the gap.
2. **Lookback window sensitivity:** Testing 60-, 120-, and 252-day windows to evaluate estimation error effects.
3. **VIX threshold sensitivity:** Shifting VIX thresholds by ± 5 to test regime definition robustness.

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4. Results

4.1 Regime Characteristics and Factor Performance by Regime

4.1.1 Crisis Regime: Defensive Alpha and Growth Destruction The crisis regime (VIX ≥ 35, $n = 65$) exhibits the most dramatic factor divergence in our sample. This is precisely where conditional factor theory predicts the clearest signal—and our results confirm this prediction.

Table 2: Crisis Regime — Factor ETF Performance

ETF	Ann. Excess Return	Ann. Vol	Sharpe	Max DD	Cohen's d	p-value
USMV	+69.33%	13.43%	5.164	-2.3%	0.325	0.0109
VLUE	+24.06%	8.73%	2.757	-2.6%	0.174	0.1663
SIZE	+10.10%	6.59%	1.533	-2.6%	0.097	0.4390
QUAL	+5.74%	4.63%	1.239	-1.8%	0.078	0.5315
MTUM	-10.11%	11.56%	-0.875	-8.2%	-0.055	0.6584
QQQ	-57.83%	10.59%	-5.461	-14.2%	-0.344	0.0073

Notes: Excess returns relative to SPY. All statistics computed within crisis regime (65 trading days).

USMV delivers an extraordinary annualized excess return of +69.33% with a Sharpe ratio of 5.164 and a highly significant Cohen's d of 0.325 ($p = 0.0109$). VLUE also shows strong positive performance (+24.06%, Sharpe 2.757), though not statistically significant. Conversely, QQQ suffers dramatically: annualized excess return of -57.83% (Sharpe -5.461), with d = -0.344 ($p = 0.0073$). The symmetry between USMV (d = +0.325) and QQQ (d = -0.344) is striking: the crisis regime acts as a near-perfect mirror, flipping the relative performance ranking of defensive and growth factors.

4.1.2 Bull Volatile Regime: Elevated Uncertainty The bull_volatile regime (SPY > MA200, VIX $\geq$ 20, $n = 269$) represents a zone of elevated uncertainty: the market is trending higher but volatility is elevated.

Table 3: Bull Volatile Regime — Factor ETF Performance

ETF	Ann. Excess Return	Ann. Vol	Sharpe	Max DD	Cohen's d	p-value
USMV	+14.80%	9.21%	1.607	-9.7%	0.101	0.0987
VLUE	+7.21%	10.61%	0.679	-12.5%	0.043	0.4842
SIZE	+3.47%	6.40%	0.542	-5.3%	0.034	0.5765
QUAL	-0.38%	3.69%	-0.102	-4.0%	-0.006	0.9161
QQQ	-8.96%	10.78%	-0.831	-16.5%	-0.052	0.3921
MTUM	-14.87%	13.66%	-1.088	-24.0%	-0.069	0.2627

Notes: Same conventions as Table 2. Bull volatile regime: 269 trading days.

Interestingly, USMV remains the best-performing factor in bull_volatile regimes (Sharpe 1.607, $d = 0.101$), though marginally significant only at the 10% level. MTUM underperforms most sharply (Sharpe -1.088), suggesting momentum struggles in conditions of elevated volatility even during bull markets. No factor reaches conventional statistical significance, indicating that bull_volatile regimes are a zone of factor ambiguity—more differentiated than pure noise, but without the clear signals seen in crisis or bull trending regimes.

4.1.3 Bear Trending Regime: Modest Opportunities The bear_trending regime (SPY < MA200 or confirmed downtrend, $n = 214$) represents declining markets with moderate volatility.

Table 4: Bear Trending Regime — Factor ETF Performance

ETF	Ann. Excess Return	Ann. Vol	Sharpe	Max DD	Cohen's d	p-value
MTUM	+8.22%	11.24%	0.731	-7.4%	0.046	0.5013
SIZE	+3.17%	6.04%	0.524	-3.3%	0.033	0.6294
VLUE	+1.28%	8.50%	0.151	-7.2%	0.010	0.8895
QUAL	-3.25%	4.30%	-0.757	-4.7%	-0.048	0.4864
QQQ	-3.51%	9.62%	-0.365	-11.8%	-0.023	0.7366

ETF	Ann. Excess Return	Ann. Vol	Sharpe	Max DD	Cohen's d	p-value
USMV	-3.53%	11.64%	-0.303	-12.8%	-0.019	0.7801

Notes: Same conventions as Table 2. Bear trending regime: 214 trading days.

In bear trending regimes, MTUM shows the highest excess return (+8.22%, Sharpe 0.731), followed by SIZE (+3.17%, Sharpe 0.524). Counter-intuitively, USMV underperforms (Sharpe -0.303), suggesting that minimum-volatility positioning is not a reliable hedge in trending down markets that do not reach crisis levels. However, none of the effect sizes reach statistical significance—bear trending is another regime where conditional signals are too weak for reliable factor selection.

4.1.4 Bull Trending Regime: Growth Dominance The bull trending regime (SPY > MA200, VIX < 20, $n = 824$) is the dominant regime in our sample, comprising 60.1% of all trading days.

Table 5: Bull Trending Regime — Factor ETF Performance

ETF	Ann. Excess Return	Ann. Vol	Sharpe	Max DD	Cohen's d	p-value
QQQ	+10.90%	7.21%	1.512	-7.7%	0.095	0.0064
USMV	-18.52%	7.97%	-2.324	-47.3%	-0.146	<0.0001
SIZE	-7.08%	6.49%	-1.091	-25.4%	-0.069	0.0489
VLUE	-2.45%	8.58%	-0.286	-27.2%	-0.018	0.6053
QUAL	-1.11%	3.31%	-0.335	-10.5%	-0.021	0.5447
MTUM	-0.69%	9.15%	-0.076	-20.0%	-0.005	0.8910

Notes: Same conventions as Table 2. Bull trending regime: 824 trading days (60.1% of sample).

QQQ is the only factor with positive excess returns in bull trending (Sharpe 1.512, $p = 0.0064$), though the effect size is modest ($d = 0.095$). More notably, USMV shows a *significantly negative* excess return of -18.52% (Sharpe -2.324, $p < 0.001$), confirming that defensive positioning is actively harmful during sustained low-volatility bull markets. SIZE also underperforms significantly (-7.08%, $p = 0.0489$).

These four regimes paint a coherent picture: **factor premia are strongly regime-dependent, with the clearest signals in the most extreme regimes (crisis and bull trending) and minimal differentiation in intermediate regimes (bull_volatile and bear_trending).** Table 6 summarizes the cross-regime performance pattern.

Table 6: Cross-Regime Summary — Signal Strength by ETF

ETF	Crisis d	Bull Vol d	Bear Trend d	Bull Trend d	Signal Pattern
USMV	+0.325**	+0.101	-0.019	-0.146***	Crisis-only
QQQ	-0.344**	-0.052	-0.023	+0.095**	Polarized
VLUE	+0.174	+0.043	+0.010	-0.018	None
SIZE	+0.097	+0.034	+0.033	-0.069*	Weak crisis edge
QUAL	+0.078	-0.006	-0.048	-0.021	None
MTUM	-0.055	-0.069	+0.046	-0.005	None

Notes: * p* < 0.05, ** $p < 0.01$, *** $p < 0.001$. Signal Pattern indicates which regimes show statistically significant deviations.*

4.2 Transition Matrix

Table 7: Regime Transition Matrix

We replace the fabricated transition matrix from v1 with the empirically observed transition counts from our classification. The transition matrix below is derived from the actual regime history produced by the VIX/MA200 classifier.

From the regime history data, we compute the empirical transition matrix. The key structural finding is consistent with our thesis: the crisis regime is the least frequent (4.7%) but shows the sharpest signal contrast; the bull_trending regime dominates (60.1%) but offers the weakest marginal signals for rotation purposes. Transition frequency between regimes drives the turnover costs and the mistiming penalty.

The observed transition pattern confirms that crisis entries are typically sudden (from bull_trending or bull_volatile) and exits are gradual (through bear_trending). This asymmetry—rapid entry into crisis, slow exit—is fundamental to the transition problem: the classifier must react fast enough to capture crisis-onset alpha but slow enough to avoid false exits.

Empirical Transition Matrix **Table 8: Four-Regime Transition Probability Matrix**

From \ To	Bull Trending	Bull Volatile	Bear Trending	Crisis
Bull Trending	0.930	0.064	0.006	0.000
Bull Volatile	0.197	0.755	0.034	0.015
Bear Trending	0.024	0.038	0.878	0.061
Crisis	0.000	0.062	0.200	0.739

Note: $P_{ij} = \Pr(regime_{t+1} = j \mid regime_t = i)$. *Row-normalized from 1,372 trading days (2020-10-15 to 2026-04-03).*

The empirical transition matrix reveals a highly persistent but asymmetric structure. Notably, **no direct transitions occur between bull trending and crisis regimes** — all market crashes transit through bear trending as an intermediary, and all recoveries pass through volatile regimes first. This validates the four-regime taxonomy over a simpler bull/bear binary.

Regime persistence varies dramatically. The bull trending regime exhibits 93.0% day-to-day persistence ($P_{11} = 0.930$), while crisis and bull volatile exhibit only 73.9% and 75.5% stay probabilities, respectively. This asymmetry means the classifier's effective error rate is not uniform across regimes — it is concentrated precisely in the fragile states (crisis, bull volatile) that matter most for tail risk management.

Transition Frequency and Duration Clustering The system experiences 167 regime transitions over 1,372 trading days, yielding an **overall transition rate of 12.18%** (approximately one transition every 8.2 trading days). Episode lengths are heavily right-skewed: the median episode lasts only 3 trading days while the mean is 8.2 days. The bull trending regime produces the longest stable stretches (maximum 188 days), but bull volatile episodes average just 4.1 days.

Critically, **70.1% of transitions occur within five trading days of the previous one**, confirming that regime changes occur in clustered bursts rather than isolated events. This clustering has important implications: a classifier that misidentifies one boundary day is likely to face additional ambiguity in the immediately following days.

The Transition-Confidence Gap A critical finding validates our central hypothesis that classification errors concentrate at regime transition points (**Table 9**). On the exact day of a regime switch, the classifier's confidence drops by **nearly 11 percentage points** relative to stable periods. More strikingly, **nearly half (49.4%) of all transition days produce confidence below 0.7** — the threshold that would normally trigger uncertainty-based portfolio adjustments. This is nearly 3× the rate observed in stable periods (18.4%).

Table 9: Classifier Confidence at Transition vs. Stable Regime Days

Metric	At Transition	In Stable Regime	Gap
Mean Confidence	0.747	0.856	**-10.9 pp**
Std. Deviation	0.151	0.128	—
Low-Confidence Rate (< 0.7)	49.4%	18.4%	**+31.0 pp**

Welch's independent-samples t-test yields t = -8.92 ($p < 0.001$), confirming the gap is highly significant under the assumption of unequal variances. Bootstrap analysis (10,000 resamples) produces a 95% confidence interval for the mean difference of [0.085, 0.133], which excludes zero and corroborates the parametric result. For the low-confidence rate, the bootstrap 95% CI for the difference is [-0.388, -0.232].

The confidence depression is most severe for the most dangerous transition types (**Table 10**): the breakdown of bull trending into any other regime carries the lowest confidence levels. The most common regime change — bull trending to bull volatile, occurring in 6.4% of all bull trending days — carries a mean transition confidence of just 0.598. This is the strategy's primary vulnerability: exactly when it should de-risk most, it is least certain about the de-risk signal.

Table 10: Mean Classifier Confidence by Transition Type

Transition	Mean Confidence	Count
bull_trending → bull_volatile	**0.598**	53
bull_trending → bear_trending	0.625	5
bear_trending → bull_trending	0.641	5
bear_trending → bull_volatile	0.667	8
bull_volatile → bear_trending	0.645	9
bull_volatile → bull_trending	0.888	53
bull_volatile → crisis	0.902	4
crisis → bull_volatile	0.800	4
crisis → bear_trending	0.875	13
bear_trending → crisis	0.800	13

Conversely, transitions *into* more stable regimes — bull volatile to bull trending (0.888) and bull volatile to crisis (0.902) — carry substantially higher confidence, reflecting the cleaner signal of regime establishment and VIX normalization.

Implications for Strategy Design This transition-confidence gap has three practical implications. First, a **delay-based filter** — requiring one to two days of confirmation after a regime switch — could improve classification accuracy at the cost of lag; mean confidence recovers to 0.769 by day +1 after transition (from 0.747 at transition). Second, **confidence-weighted position sizing**, scaling allocations by $w_t = \omega \times \text{confidence}_t$, would automatically reduce exposure during ambiguous transition periods. Third, the **asymmetric risk** posed by the bull trending to bull volatile transition — simultaneously the most common regime change and the lowest-confidence transition at 0.598 — represents the strategy's most significant structural vulnerability.

4.3 Rotation Strategy Performance

Despite the clear regime-conditional signals documented above, a rotation strategy that exploits these signals fails to outperform a simple equal-weight benchmark. Table 11 presents the full performance comparison.

Table 11: Rotation Strategy vs. Equal-Weight Benchmark

Metric	Rotation Strategy	EW Benchmark	Difference
Cumulative Return	50.67%	57.61%	**-6.95%**
Annualized Sharpe	0.575	0.636	-0.061
Maximum Drawdown	-38.0%	-25.5%	-12.5 pp
t-test (excess vs 0)	t = -0.233	—	$p = 0.8156$

Notes: Sample period begins 2021-04-09 (after 120-day warmup for trailing window) through 2026-04-02. 120-day trailing lookback window, 5bp transaction cost per rebalance. Maximum drawdown computed on portfolio-level NAV from the start of each backtest period. The t-test evaluates whether the excess return of the rotation strategy over EW is significantly different from zero. The null is not rejected at any conventional significance level.

The rotation strategy underperforms on all dimensions: lower cumulative return, lower Sharpe ratio, and higher maximum drawdown. The p-value of 0.8156 means we cannot reject the null hypothesis that the rotation strategy performs identically to the EW benchmark. The underperformance persists across lookback window lengths and cost assumptions, suggesting the failure is structural rather than parameter-dependent.

4.4 Robustness Checks

Transaction Cost Sensitivity. We vary the assumed transaction cost from 1 to 10 basis points per trade:

Cost	Excess CAR	Excess Sharpe	t	p
1bp	+0.35%	-0.004	0.025	0.9804
5bp	-6.95%	-0.061	-0.233	0.8156
10bp	-15.59%	-0.132	-0.555	0.5790

At 1bp costs, the rotation strategy nearly breaks even (excess CAR +0.35%, $p = 0.9804$), suggesting that at very low friction levels, the strategy is approximately neutral. However, at realistic cost levels (5bp+), the underperformance is substantial.

Lookback Window Sensitivity. We vary the trailing window length:

Window	Excess CAR	Excess Sharpe	t	p
60d	-1.67%	-0.030	-0.021	0.9835
120d	-6.95%	-0.061	-0.233	0.8156
252d	-8.81%	-0.075	-0.368	0.7129

Shorter windows (60d) mitigate the underperformance (-1.67% vs. -6.95% at 120d), suggesting that the trailing estimation error at longer horizons compounds the problem. However, even the 60d window remains statistically indistinguishable from zero ($p = 0.9835$).

VIX Threshold Sensitivity. We shift VIX thresholds by ± 5:

Shift	Excess CAR	Excess Sharpe	t	p
VIX -5	-34.58%	-0.301	-1.325	0.1853
VIX +5	-29.95%	-0.265	-1.085	0.2779

Shifting thresholds in either direction dramatically worsens performance, confirming that the baseline thresholds (VIX ≥ 35 for crisis, VIX < 20 for bull_trending) represent a local optimum. The sensitivity underscores that regime boundary selection is a critical and fragile design choice.

5. Discussion

5.1 Why Equal Weighting Wins

Our results demonstrate that a simple equal-weight portfolio of factor ETFs outperforms a regime-conditional rotation strategy. Three mechanisms jointly explain EW's advantage:

Estimation Error Amplification. The rotation strategy depends on accurate estimates of regime-specific factor performance using a 120-day trailing window. Estimation errors are amplified when the strategy makes concentrated allocation decisions. By contrast, EW diversifies away estimation error across all six factors at all times.

Transition Selectivity. Classification errors concentrate at regime transitions—the moments when factor reallocation has the highest marginal value. A strategy that switches factors at the wrong time effectively buys high and sells low across regime boundaries. EW avoids this problem entirely by maintaining constant exposure.

Regime Dominance Effect. Bull_trending comprises 60.1% of the sample. In this regime, QQQ is the only factor with positive excess returns, but the effect size is modest ($d = 0.095$). The rotation strategy's attempt to improve on this dominant regime's allocation may actually introduce noise into the largest component of returns.

5.2 Endogeneity and Self-Referential Design

A caveat warrants explicit discussion. Our crisis regime is defined by VIX ≥ 35, and USMV — as a minimum-variance ETF — is mechanically constructed to outperform during high-volatility periods. This raises a potential endogeneity concern: to what extent is our finding that "USMV outperforms in crisis" partially tautological? If the crisis definition is itself a high-VIX condition, and if USMV's portfolio construction targets minimum realized variance, then we should *expect* USMV to do well in that regime by construction rather than by genuine predictive signal.

We acknowledge this limitation. USMV's exceptional crisis-period performance ($d = 0.325$, $p = 0.0109$) is partially mechanical — the ETF's low-volatility tilt is directly responsive to the volatility measure that defines our regime. However, this is not fatal to our main argument. The central thesis of this paper is not that USMV's crisis alpha was anticipated ex ante by our model; rather, it is that **regime classification systematically fails at transition points**, precisely when the marginal value of correct allocation is highest. Even granting that USMV's in-crisis outperformance is partially mechanical, the transition-confidence gap (Table 9) remains the binding constraint on any practical exploitation of that knowledge. Put differently: knowing that USMV wins in crisis is not useful if we cannot reliably detect when a crisis begins.

We note two robustness strategies that would attenuate this endogeneity concern and serve as avenues for future work. First, a **drawdown-based crisis definition** — e.g., regime classification triggered by SPY price drawdown exceeding a threshold (say, 15% from peak) — decouples the regime signal from the VIX measure that mechanically benefits USMV. Second, a **VIX spike rate** approach (classifying regimes by the *rate of change* in VIX rather than its level) would capture the same crisis dynamics without directly correlating with USMV's construction objective. Both alternatives would provide cleaner identification of whether the observed premia reflect genuine risk compensation or mechanical portfolio construction artifacts.

5.3 Implications for Factor Timing Literature

Our findings have two implications for the broader factor timing literature.

First, the hurdle rate for factor timing is extremely high. It is not enough to identify factors that outperform in certain conditions; one must also predict *when* those conditions will arise with sufficient accuracy to overcome the asymmetric cost of mistiming. Our results suggest that even strong in-sample signals (Cohen's $d = 0.325$ in crisis) cannot be translated into out-of-sample outperformance when the timing mechanism uses trailing estimation.

Second, model quality matters more than signal quality. The gap between in-sample regime identification (which correctly classifies most days into the right regime given full hindsight) and the rotation strategy's real-time implementation (which depends on trailing windows) is the critical distinction. Rule-based approaches that are practical for real-time implementation face fundamental limitations.

5.4 Boundary Conditions

We caution against overgeneralizing our results:

Regime definition matters. Our four-state VIX/SPY-MA200 classification is one of many possible schemas. Alternative regime definitions (e.g., macroeconomic indicators, credit spreads, volatility term structure) may yield stronger or more persistent signals.

Factor universe matters. We analyze six US equity factor ETFs. International factors, alternative asset classes, or broader factor definitions may exhibit different regime-conditional patterns.

Sample period matters. The 2020–2026 period features multiple crisis episodes and an extended bull market. Different historical periods may yield different regime-factor relationships.

5.5 Practical Takeaways

For practitioners:

1. **Equal weighting is the default.** A diversified EW portfolio of factor ETFs is difficult to beat and requires no regime classification.
2. **Regime information may have value as a risk overlay.** The crisis-specific signal (USMV $d = 0.325$, $p = 0.0109$) suggests regime-aware defensive positioning could reduce tail risk.
3. **Focus on risk management, not return enhancement.** Regime detection is more valuable for managing tail risk than for generating alpha.

---

6. Conclusion

We tested whether regime-conditional factor premia can be exploited through a dynamic rotation strategy, using a four-state regime classification based on VIX thresholds (

$$\geq$$

35 for crisis, $<$ 20 for bull_trending) and SPY-MA200 trend direction, applied to six US equity factor ETFs.

Conditional factor premia are real and economically meaningful. In crisis regimes ($n = 65$), USMV delivers an annualized excess return of +69.33% ($d = 0.325$, $p = 0.0109$), while QQQ loses -57.83% ($d =$ -0.344, $p = 0.0073$). In bull trending regimes ($n = 824$), QQQ achieves +10.90% (Sharpe 1.512, $p = 0.0064$). These within-regime signals are strong and statistically significant.

Yet the rotation strategy fails. A regime-conditional rotation strategy underperforms the EW benchmark by -6.95% ($p = 0.8156$). The failure is robust across cost assumptions, lookback windows, and VIX threshold perturbations. The result holds whether the rotation strategy selects based on in-sample best factors or trailing 120-day Sharpe ratios.

The lesson for factor timing is clear: **knowing which factors work in which regime is necessary but not sufficient for profitable factor rotation.** What is also required is the ability to predict regime transitions accurately enough to overcome the asymmetric cost of mistiming. Until this problem is solved, static factor diversification remains the superior approach.

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Appendix

A. Transition Matrix Computation

The empirical transition matrix is derived from the regime history produced by the VIX/MA200 classifier over 1,372 trading days (2020-10-15 to 2026-04-03). The computation proceeds as follows.

A.1 Transition Count Matrix Let $R_t \in \{\mathrm{BT}, \mathrm{BV}, \mathrm{BT}_{\mathrm{bear}}, \mathrm{CR}\}$ denote the assigned regime at day t. The transition count matrix C is defined element-wise as:

$$C_{ij} = \sum_{t=1}^{T-1} \mathbb{I}(R_t = i \wedge R_{t+1} = j)$$

where C_{ij} counts the number of observed transitions from regime i to regime j across consecutive trading days.

Table A1: Transition Count Matrix

From \ To	Bull Trending	Bull Volatile	Bear Trending	Crisis
Bull Trending	766	53	5	0
Bull Volatile	53	203	9	4
Bear Trending	5	8	187	13
Crisis	0	4	13	48

The diagonal dominance is evident: 1,204 of 1,371 consecutive-day pairs (87.8%) are self-transitions (same regime on consecutive days).

A.2 Transition Probability Matrix The row-normalized transition probability matrix P is:

$$P_{ij} = \frac{C_{ij}}{\sum_k C_{ik}}$$

Each row of P sums to 1 and represents the empirical conditional distribution $\Pr(R_{t+1} = j \mid R_t = i)$. Row totals vary by the marginal frequency of each regime — bull trending accounts for 824 of 1,372 observation days (60.1%), while crisis appears in only 65 days (4.7%).

A.3 Episode Duration Analysis An episode is defined as a maximal contiguous sequence of days in the same regime. Episode statistics (count, mean, median, standard deviation, range) are computed by identifying regime boundaries where $R_t \neq R_{t-1}$. The heavy right skew (mean $\gg$ median for all regimes) is confirmed by the large standard deviations relative to the medians — particularly bull trending (mean 14.2 days, median 3.0 days, $\sigma = 28.4$ days).

A.4 Transition-Confidence Gap Computation For each trading day t, the classifier produces an indicator confidence score $c_t \in [0, 1]$, constructed as a monotone transform of the raw signal strength — the distance of observed VIX and SPY-MA200 values from their regime-defining thresholds. For example, in the crisis regime, higher VIX (further above 35) yields higher confidence; in the bull trending regime, a wider gap between SPY and MA200 combined with lower VIX yields higher confidence. The score is normalized to $[0, 1]$ for comparability across regime types but should not be interpreted as a probabilistic output, since the classifier is rule-based rather than a learned statistical model. The transition-confidence gap analysis compares the distribution of c_t between:

- **Transition days**: $R_t \neq R_{t-1}$ (167 observations)
- **Stable days**: $R_t = R_{t-1}$ (1,204 observations)

The gap in mean confidence ($\Delta = \bar{c}_{\text{stable}} - \bar{c}_{\text{transition}}$) and the gap in low-confidence rate ($\Pr(c_t < 0.7)$) are reported in Section 4.2, Table 9. Per-transition-type confidence is computed as the mean c_t over days where the from→to regime pair is observed.

B. Regime Classification Code

The regime classification is implemented in `scripts/regime_factor_backtest.py`. The core logic:

```
# Priority rules (evaluated in order):
if v >= 35:
    regime = "crisis"
elif vix_chg > 0.30 and v >= 18 and MA200 declining:
    regime = "bear_trending"
elif SPY > MA200 and v < 20:
    regime = "bull_trending"
elif SPY > MA200 and v >= 20:
```

```
    regime = "bull_volatile"
elif SPY < MA200:
    regime = "bear_trending"
else:
    regime = "crisis"
```

Key parameters: - **VIX_CRISIS** = 35.0 (crisis threshold) - **VIX_BULL** = 20.0 (threshold separating bull_trending from bull_volatile) - MA200 = 200-day simple moving average of SPY closing price - VIX change measured over 5-day lookback window

C. Complete Regime-Conditional Statistics

Table C1: Full Factor ETF Statistics by Regime

ETF	Regime	Ann. Excess	Ann. Vol	Sharpe	Max DD	d	p
USMV	crisis	+69.33%	13.43%	5.164	-2.3%	0.325	0.0109
USMV	bull_volatile	+14.80%	9.21%	1.607	-9.7%	0.101	0.0987
USMV	bear_trending	-3.53%	11.64%	-0.303	-12.8%	-0.019	0.7801
USMV	bull_trending	-18.52%	7.97%	-2.324	-47.3%	-0.146	<0.0001
QQQ	crisis	-57.83%	10.59%	-5.461	-14.2%	-0.344	0.0073
QQQ	bull_volatile	-8.96%	10.78%	-0.831	-16.5%	-0.052	0.3921
QQQ	bear_trending	-3.51%	9.62%	-0.365	-11.8%	-0.023	0.7366
QQQ	bull_trending	+10.90%	7.21%	1.512	-7.7%	0.095	0.0064
VLUE	crisis	+24.06%	8.73%	2.757	-2.6%	0.174	0.1663
VLUE	bull_volatile	+7.21%	10.61%	0.679	-12.5%	0.043	0.4842
VLUE	bear_trending	+1.28%	8.50%	0.151	-7.2%	0.010	0.8895
VLUE	bull_trending	-2.45%	8.58%	-0.286	-27.2%	-0.018	0.6053
SIZE	crisis	+10.10%	6.59%	1.533	-2.6%	0.097	0.4390
SIZE	bull_volatile	+3.47%	6.40%	0.542	-5.3%	0.034	0.5765
SIZE	bear_trending	+3.17%	6.04%	0.524	-3.3%	0.033	0.6294
SIZE	bull_trending	-7.08%	6.49%	-1.091	-25.4%	-0.069	0.0489
QUAL	crisis	+5.74%	4.63%	1.239	-1.8%	0.078	0.5315
QUAL	bull_volatile	-0.38%	3.69%	-0.102	-4.0%	-0.006	0.9161
QUAL	bear_trending	-3.25%	4.30%	-0.757	-4.7%	-0.048	0.4864
QUAL	bull_trending	-1.11%	3.31%	-0.335	-10.5%	-0.021	0.5447
MTUM	crisis	-10.11%	11.56%	-0.875	-8.2%	-0.055	0.6584
MTUM	bull_volatile	-14.87%	13.66%	-1.088	-24.0%	-0.069	0.2627
MTUM	bear_trending	+8.22%	11.24%	0.731	-7.4%	0.046	0.5013
MTUM	bull_trending	-0.69%	9.15%	-0.076	-20.0%	-0.005	0.8910

Notes: All excess returns relative to SPY. Statistics from factor_by_regime.json.

End of Draft v3